R16

Code No: 138EK

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations July - 2023
REAL-TIME SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper July/Aug-2023

PART – A

(25 Marks)

1.a)	What is fork and write command?	[2]
b)	What is the role of kernel?	[3]
c)	List the basic design principles in RTOS.	[2]
d)	Explain the importance of queues in RTOS.	[3]
e)	What is a Register?	[2]
f)	Define Dead lock.	[3]
g)	Determine the cause of the interrupt.	[2]
h)	What are the different types of exceptions?	[3]
i)	What is Vx works?	[2]
j)	What are the main challenges of choosing RTOS for an embedded system design?	[3]

PART – B

(50 Marks)

2.a) Write a Shell script that displays the number of readable, writable, and executable files in a specific directory.
b) Explain the file I/O commands briefly. [5+5]

OR

3.a) Explain any five Linux general file manipulation commands with examples.
b) Explain about common shells used in UNIX systems. [5+5]

4. Explain how Round-Robin scheduling works with an illustrative example in RTOS.[10]

OR

5.a) Explain about the different types of real time operating systems with examples.
b) Describe semaphore functions for inter task communication. [5+5]

6.a) What is meant by a pipe? How does a pipe differ from a queue?
b) Briefly explain about performance metrics for RTOS. [5+5]

OR

7.a) Give a detailed description of signal control block and explain various operations of it.
b) Give examples of I/O subsystems. Explain the use of asynchronous I/O. [5+5]

8.a) Describe in detail about spurious interrupts and how to handle them.
b) List the three ways in which an RTOS handles the ISRs in a multitasking environment. [5+5]

OR

9.a) Discuss about Interrupt Routines in RTOS Environment.
b) What are exceptions and interrupts? Explain the applications of interrupts. [5+5]

10.a) Explain the inheritance protocol implementation in RT Linux.
b) Explain the Case Study of Inter-Robot Communication in a Robot Orchestra. [5+5]

OR

11.a) Explain the basic features of RTOS VxWorks.
b) Explain the Case Study of Appliance Control by RT Linux System. [5+5]

---ooOoo---

Used paper July/Aug-2023